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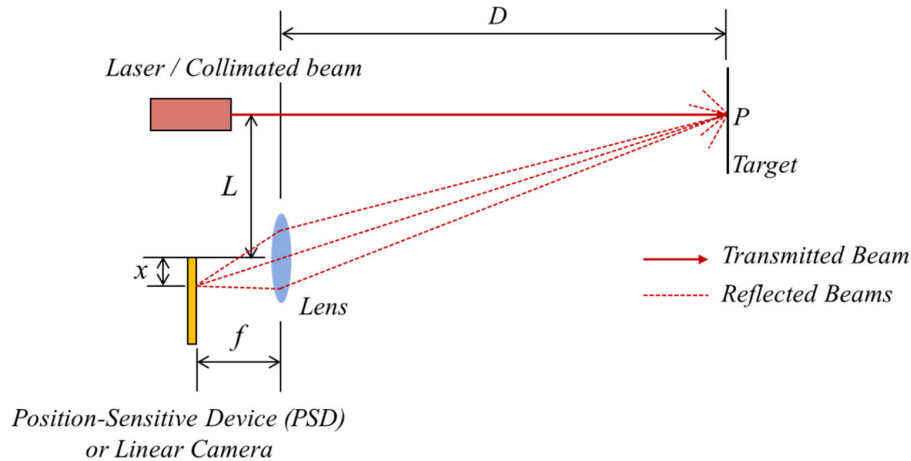
Exercise 4: Sensor Models

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Task 1: Triangulation Sensor

Consider the laser triangulation sensor shown below. A laser beam is pointed at the target and diffusely reflected through the lens onto a position-sensitive device (PSD). For the questions involving uncertainty, you may use the general, multi-dimensional formulation of the error propagation law

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \mathbf{y} = \mathbf{f}(\mathbf{x}) \stackrel{\text{approx.}}{\sim} \mathcal{N}\left(f(\boldsymbol{\mu}), \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\boldsymbol{\mu}} \boldsymbol{\Sigma} \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\boldsymbol{\mu}}^{\top}\right).$$



- i) Find the direct measurement model as introduced in the lecture

$$z_x = h_x(D) + v_x, \quad v_x \sim \mathcal{N}(0, \sigma_x^2).$$

- ii) In practice, we are interested in the depth. Invert the model and write the distance measurement as

$$z_D = h_x^{-1}(D) + v_D = h_D(x) + v_D, \quad v_D \sim \mathcal{N}(0, \sigma_D^2).$$

- iii) Sketch the direct model, i.e. z_x as a function of D , assuming $L = 1$ and $f = 1$. Based on the sketch, discuss how the inversion affects the measurement quality (resolution, sensitivity to noise) for targets that are close and far away.
- iv) We are now interested in the uncertainty of the inverse model. Assume that the measurement $z_x \sim \mathcal{N}(x, \sigma_x^2)$ where x would be the value of the ideal, noise-free measurement. Using linear error propagation and the inversion function derived in subtask ii), find the uncertainty σ_D of the inverse model $z_D \sim \mathcal{N}(h_D(x), \sigma_D^2)$. Note that as we are dealing with a one-dimensional measurement, the formulation of the error propagation law reduces to a scalar equation.
- v) Sketch σ_D against the measured distance D assuming $\sigma_x = 0.1$. What does this equation tell you about the uncertainty for targets that are far away?

Task 2: Laser ToF (Time of Flight) Distance Sensors on Diff-Drive Robot (Python)

Consider a two-wheel differential drive mobile robot with the following states:

$$\mathbf{x}^T = [t_x, t_y, \theta]^T,$$

where t_x, t_y denote the 2D position of the robot in the world frame, and θ is the orientation of the robot frame relative to the world frame. Now the robot has three laser ToF distance sensors (index $i \in 1, 2, 3$), mounted at the robot body frame origin, pointing left ($i = 1$), forward ($i = 2$) and right ($i = 3$). Assume their measurement range is $d_{\min}$ to $d_{\max}$. While driving around in a room, it will now take distance measurements to the surrounding walls.

For installation of required packages, we provide you with a requirements.txt. You can install them using e.g. `pip install -r requirements.txt`

- i) Open the Python Script `e04_diff_drive.py` to see the set-up: the simulation will show the robot carrying out a 2D motion in a room.
- ii) Develop the ideal, noise and bias-free measurement model for the i^{th} sensor $d_i = h_i(\mathbf{x})$ by considering the intersection of the sensor ray and the wall in the world frame. Assume that the parameters a_1, a_2, a_3 of the line equation $a_1x + a_2y + a_3 = 0$ are known: these correspond to the correctly identified wall segment that the sensor is pointing at.
- iii) Now a full probabilistic measurement model is developed on top of this. Assume that we have additive zero-mean Gaussian noise that affects the measurement with a standard deviation of σ . With a small probability of P_{outlier} , the sensor will return a nonsensical result uniformly distributed within the measurement range. Write down the full measurement model for d_i and draw the probability density function for the measurement $p(d_i)$.
- iv) Now implement the measurement function into the Python script `e04_diff_drive.py` that is used to simulate measurements. The function should include correctly selecting the wall segment, and generate a sample from the probabilistic model above. If the distance is out of measurement range, the sensor should simply not return a measurement, i.e. an empty list. Furthermore, plot the measurements into the world frame to see, qualitatively, if things make sense. The necessary numeric values for σ , $d_{\min}$, $d_{\max}$, and P_{outlier} are already defined, but you can obviously play with them and check the effects.